



## Review

# Monitoring cognitive function in the fatigued warfighter: A rapid review of cognitive biomarkers



Luana C. Main <sup>a,\*</sup>, Larisa T. McLoughlin <sup>b</sup>, Shawn D. Flanagan <sup>c</sup>, Maria C. Canino <sup>c</sup>, Siobhan Banks <sup>b</sup>

<sup>a</sup> Institute for Physical Activity and Nutrition (IPAN), School of Exercise and Nutrition Sciences, Deakin University, Australia

<sup>b</sup> Behaviour-Brain-Body Research Centre, University of South Australia, Australia

<sup>c</sup> Neuromuscular Research Laboratory, University of Pittsburgh, United States

## ARTICLE INFO

## Article history:

Received 5 June 2022

Received in revised form 27 April 2023

Accepted 27 April 2023

Available online 6 May 2023

## Keywords:

Attention

Cognition

Memory

Sleep

Soldiers

Task performance

## ABSTRACT

**Objectives:** Decreases in cognitive function impair occupational performance, reduce occupational safety, and increase musculoskeletal injury risk. The aim of this paper was to identify measures that may be used to monitor cognitive function in the warfighter.

**Design:** A rapid review.

**Methods:** A rapid search of Academic Search Complete, MEDLINE, PsycINFO, and SPORTSDiscus databases was conducted. Eligibility criteria: original peer reviewed research articles, written in English, published between 2002 and 2022, and using human participants with no health issues in military training environments or active service.

**Results:** Of the 248 articles screened, 58 full-text articles were assessed for eligibility and 29 included in the review. Of these, 16 papers presented data from multi-stressor military training environments, or experimental studies where simulated military tasks were being performed. Thirteen papers focused on an aspect of military work and the implications for cognitive function (i.e., physical load, periods of extended wakefulness or fatigue, and hypoxic conditions). The domains of cognitive function that were assessed (i.e. vigilance, reaction time, working memory, situational awareness, and decision-making) were somewhat consistent among studies.

**Conclusions:** Prolonged exposure to high-stress military environments compromises multiple aspects of cognitive function. These findings highlight the need for a suite of biomarkers to monitor cognitive function and assess the ability of military personnel to attend to and process mission-critical information and make appropriate decisions on the battlefield and other high-stress environments. Our findings suggest that a suite of common tests may provide useful information about cognitive function in the warfighter.

© 2023 The Author(s). Published by Elsevier Ltd on behalf of Sports Medicine Australia. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

## Practical implications

- Declines in cognitive function may be indicative of sleep deprivation or hypoxic conditions.
- Stress associated with sleep loss, cognitive or physical exertion, and military occupational tasks compromise cognitive function.
- Individuals are not always aware of decreases in cognitive function, and the ability to accurately self-assess cognitive function is diminished during high stress situations.
- Although motivation or working together in teams may temporarily buffer some of the negative effects of sleep loss in trained

individuals, repeated exposure with inadequate recovery will likely impair performance.

- Measures of cognitive function appear sensitive to military operational stress and viable in field settings.

## 1. Introduction

In military environments it is essential to maintain cognitive function despite heightened stress and fatigue. Fatigue is a complex state with physiological and psychological determinants, in which there is a perception that greater effort may be required to complete a given task. However, there are some important differences between cognitive fatigue and physical fatigue. From a subjective point of view, cognitive fatigue is associated with increased feelings of tiredness, reduced motivation, and a perceived lack of energy. Often, cognitive fatigue is associated with sleep loss (due to disruption, restriction, or deprivation), circadian rhythm disruption, or prolonged stress exposure. In

\* Corresponding author.

E-mail addresses: [luana.main@deakin.edu.au](mailto:luana.main@deakin.edu.au) (L.C. Main),

[larisa.mcloughlin@unisa.edu.au](mailto:larisa.mcloughlin@unisa.edu.au) (L.T. McLoughlin), [sdf29@pitt.edu](mailto:sdf29@pitt.edu) (S.D. Flanagan), [mcc132@pitt.edu](mailto:mcc132@pitt.edu) (M.C. Canino), [Siobhan.banks@unisa.edu.au](mailto:Siobhan.banks@unisa.edu.au) (S. Banks).

comparison, physical fatigue is generally considered as a function of physical workload or exertion, although cognitive factors are also consequential. In both instances, there is evidence of compromised cognitive function.

Cognitive function is a broad term that refers to multiple interrelated global and specific mental processes (or domains). There are several ways to conceptualize cognitive function, one of which is a hierarchical structure based on the complexity of the operations. Using this lens, basic sensory and perceptual processes are the least complex, or lower-level mental processes that allow a person to detect a stimulus that occurs in one of the five sensory modalities. Likewise, the execution of motor skills or the basic elements of motor activity require minimal cognitive demands. In comparison, reasoning and problem-solving, often referred to as executive functioning, require the coordination of multiple sensory, perceptual, and/or attentional processes. The ability to perform these types of higher-level processing is central to military roles where, decision-making, calculations, and language skills are required.

1.1. Consequences of decreases in cognitive function

Numerous behavioral and environmental stressors reduce cognitive function in military service members. Unique to this population, the stress caused by suboptimal sleep and prolonged cognitive or physical exertion rarely occurs in isolation and is likely compounded by the situational uncertainty, time-sensitivity, and hazardous conditions that typify military operations. Sleep disruption and restriction are common features of military operations that can lead to impairments in attentional, memory, emotional, and higher-level cognitive functions.<sup>1</sup> These cognitive impairments decrease performance on occupational tasks that rely on rapid decision-making. For instance, during simulated marksmanship tasks, after 72 h of sleep restriction, military personnel display slower reaction times and less ability to discriminate friend from foe.<sup>2</sup> Individuals' self-awareness of such deficits appears limited, which highlights the danger that suboptimal sleep poses to operational safety. The effect of sleep loss on musculoskeletal function is less clear and appears to differ by activity, but impairments in aerobic capacity and local muscular endurance are probable.<sup>3</sup> In military settings, reduced sleep ( $\leq 4$  h) coincides with a 2.4-fold increase in musculoskeletal injury incidence.<sup>4</sup> Similarly, in active-duty U.S. Army Service Members, a sleep-related medical diagnosis is associated with an increased risk of motor vehicle accidents, work/duty-related injuries, and the corresponding likelihood of being nondeployable.<sup>5</sup> Additional work is needed, but it is also likely that suboptimal sleep and overexertion contribute to psychophysiological stress.

Regardless of the cause(s), the consequences of cognitive impairment are not restricted to cognitive function but include detrimental effects on occupational performance and safety, musculoskeletal function, injury risk, and recovery. As such, efforts to maintain and improve cognitive function have important practical implications for warfighter efficacy, health, and safety in operational and non-deployed settings. The specific mental processes (or domains) of cognitive function are typically assessed with computerized tests, surveys, or tasks, the results of which can be considered as "biomarkers" (e.g. reaction time as a biomarker of attentional function, recall accuracy as a biomarker of memory function). Therefore, the aim of this paper was to identify measures that are available to monitor cognitive function in the warfighter. From a practical standpoint, given the limited ability to implement invasive (e.g., blood) and laboratory-based techniques (e.g., imaging, high fidelity sensors) in field settings, such cognitive measures may represent an efficient way to non-invasively obtain useful information about warfighter health, recovery, and readiness in military operational environments.

2. Methods

While systematic reviews are considered the gold standard for knowledge synthesis, they are not always possible due to the amount

of time required. As such, a rapid review was conducted for the current paper, which can be considered as a type of knowledge synthesis in which components of the systematic review process are simplified or omitted to produce information in a much shorter time period (i.e.,  $\leq 4$ –6 weeks). For the current paper, a systematic search of the electronic databases Academic Search Complete, MEDLINE, PsycINFO, and SPORTSDiscus was conducted. The search strategy is presented in Table 1.

2.1. Selection criteria

Databases were searched from 2002 to January 31, 2022 and papers were required to meet the following criteria: 1) original peer reviewed research articles, written in English using participants from active military or defense populations (i.e., currently serving, not veterans); contain 2) a measure of cognitive function; and 3) participants were in military training environments, active service or performing simulated military tasks, and 4) had no underlying health concerns or medical conditions (i.e., anxiety, depression, or PTSD) that might compromise cognitive function or sleep. The initial search yielded 248 results that the first author (LCM) screened for inclusion by title and abstract before progressing to full text assessment by the first, and second authors (LTM). The final studies selected for inclusion in the review were approved by all co-authors against the inclusion criteria, who were also consulted as needed to provide a second opinion on whether a study met the inclusion criteria. The screening process is described in Fig. 1.

3. Results

Fifty-eight full text manuscripts were carefully reviewed for inclusion in the results of this paper. The following points were examined: study type (lab or field study); cognitive functions studied (e.g., memory, language, executive function, attentional, perceptual, and psychomotor functions); methods used for data collection; analyses conducted; key results; practical implications; and study limitations. Noting that the aim of this paper was to identify what measures are available to monitor cognitive function in the warfighter, if studies were conducted in strict lab conditions with no immediate 'real world' applicability or relevance to current military contexts, they were excluded from the synthesis ( $n = 14$ ). The final number of papers included in the review and extracted for discussion was 29 (these are summarized in Table 2). Sixteen papers presented data from military training environments or simulated military operational stress (SMOS) scenarios.<sup>6–21</sup> Recognizing that data collection in field and active-duty environments is inherently challenging, it was felt that these simulated scenarios offer valuable insights into the effects of these situations on cognitive function. These papers will collectively

Table 1  
Database search strategy.

| Search # | Search terms                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        | military OR soldier* OR "defence force" OR "armed forces" OR navy OR airforce* OR "servicem?n" OR "servicewom?n" OR warfight*                                                                                                                                                                                                                                                                                                                                                           |
| 2        | "cognitive function" OR cognition OR perception OR memory OR learning OR "decision making" OR "mental abilities" OR "brain function" OR "working memory" OR memory OR "cognitive flexibility" OR "mental operations" OR academic OR "academic success" OR thought OR "thought processes"                                                                                                                                                                                                |
| 3        | fatigue* OR *load OR sleep OR recovery OR "under recovery" OR "sleep deprivation" OR "deprived sleep" OR "sleep restriction" OR "restricted sleep" OR "sleep disruption" OR "disrupted sleep" OR "sleep inadequa*" OR "inadequate sleep" OR "sleep quality" OR "quality sleep" OR "sleep extension" OR "extended sleep" OR "sleep disturbance" OR "disturbed sleep" OR "sleep enhancement" OR "enhanced sleep" OR "sleep duration" OR "sleep efficien*" OR "sleep loss" OR "lost sleep" |
| 4        | Search 1 AND Search 2 AND Search 3                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

Limits: full text, 2002–2022, English language, academic peer-reviewed journals, current warfighters (not veterans), no underlying health concerns or medical conditions (i.e., anxiety, depression, or PTSD) that might compromise cognitive function or sleep.

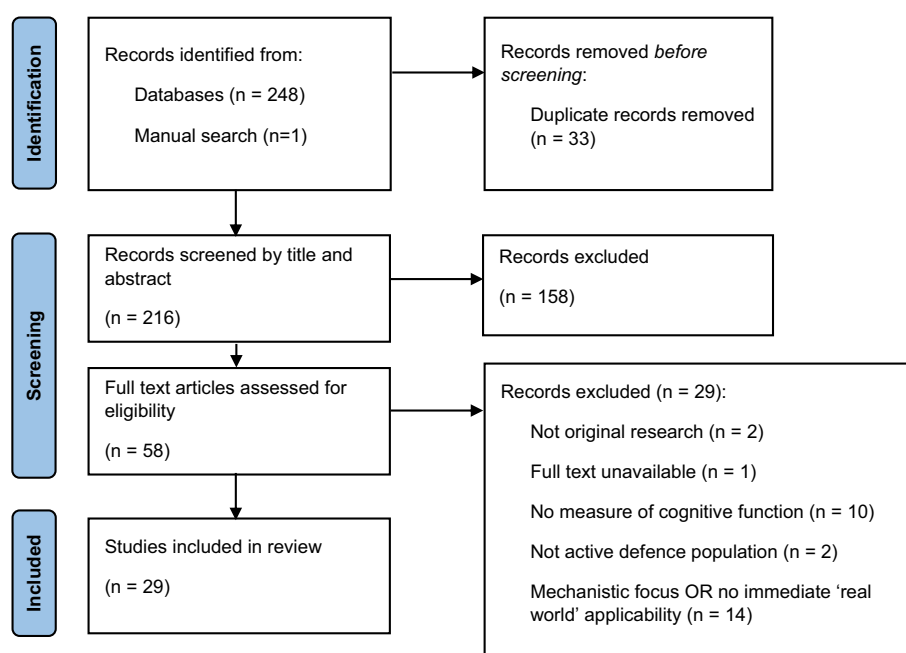


Fig. 1. Screening process for selection of studies in relation to inclusion and exclusion criteria.

be referred to as the SMOS papers in recognition of the multi-source stressors that participants may have been exposed to (including but not limited to physical loads/fatigue, sleep or caloric restrictions). Five papers investigated the effects of fatigue on various aspects of performance.<sup>2,22–25</sup> Four papers investigated the effects of physical load and exertion on task performance, and/or cognitive function.<sup>26–29</sup> Three papers examined the impact of hypoxia on cognitive function during four to six hour simulated military flights.<sup>30–32</sup> Finally, one paper examined the visual perceptual, affective and cognitive implications of equipping temporary military shelters with either fluorescent or one of three advanced LED lighting systems with varied color temperature and luminance.<sup>33</sup>

Of these studies, two investigated the ergogenic effects of caffeine on task performance,<sup>8,24</sup> and one examined the effect of carbohydrate (CHO) administration on recovery following the mock captivity phase of survival training.<sup>14</sup> Caffeine has stimulant-like effects when administered in moderate doses, improving alertness and performance on tasks that contain a vigilance component. Caffeine ingestion may also improve some cognitive tasks while sleep deprived. Similarly, participants who ingested CHO the night prior to a recovery sleep exhibited significantly improved performance on a complex cognitive task. These results suggest that cognitive function in SMOS can be augmented as needed for short periods of time with dietary supplements or ergogenic aids.

When reviewing the full texts, it was promising that there was consistency among some of the measures and types of tests used to assess cognitive function. These tests covered a range of functional domains, including vigilance, decision-making, executive functioning (e.g., planning, judgment, flexibility, problem-solving), attention, perception, and memory. One of the challenges in categorizing the cognitive tests is that most, if not all, assess multiple aspects of cognitive function. These have been summarized in Table 2. For those readers interested in the specific tests of cognitive function used in each of the reviewed studies, the authors would like to direct them to the respective manuscripts for more detail.

A number of the studies focused on specific forms (i.e., survival training)<sup>12–14,19</sup> or aspects (i.e., load carriage)<sup>27,29</sup> of military training and found reductions in cognitive function after the completion of training activities.<sup>11,16–18,20</sup> Although the majority of studies administered cognitive tests via computers, it is possible that these tests

could be used to monitor cognitive performance in the field. There were instances where it was possible to administer assessments by pen and paper, phone, or tablet device, each of which may facilitate implementation in field environments. With the exception of a few studies, the findings from the review indicate that cognitive function was sensitive to stressor exposure during military training and SMOS.

#### 4. Discussion

Military personnel are trained to tolerate high levels of psychophysiological stress inextricably linked to the hazards, uncertainty, and time sensitivity of military operations. Understanding the impact of exposure to these demanding environments and their impact on cognitive function has important implications for warfighter health, deployment strength, and combat effectiveness. As such, the aim of this review was to identify tests that could be used to assess cognitive functions in the warfighter. Acknowledging the downselection of papers, only studies that presented immediate 'real world' applicability with recruits or military personnel performing military tasks (or simulations of) were included in the review. From these, there were several studies that investigated the impact of various survival training courses on different aspects of cognitive function.<sup>12–14,19</sup> A strength of these studies was the use of parallel measures (e.g., blood-borne biomarkers) to characterize the nature or severity of environmental stress. Overall, our review indicates that tests of memory, attention, and decision-making may provide useful information about cognitive function, fatigue, and stress in the warfighter.

Whether military stress is the result of sleep disruption or restriction, caloric deficits, physically arduous work, thermal effects, or a combination of these factors; the consequences affect higher-level, attentional, and memory cognitive functions, among others. An important observation from these studies was that although partial recovery of cognitive function may occur immediately after training ends, the negative effects of psychophysiological stress exposure extend to multi-organ systems and functions outside of the brain. Consequently, some psychophysiological stress responses persist beyond the duration of stressful activity,<sup>12</sup> and recovery can occur at different rates.<sup>12,20</sup> Moreover, complete recovery may take several days to weeks.

**Table 2**  
Full text rapid review summary of included articles.

| Author (year)                          | Participants and context                                                                                          | Cognitive function                                                        | Cognitive tests                                                                        | Other measures                                                                                                                        | Key results                                                                                                                                                                                                                                                       | Practical considerations                                                                                                                                                               |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Baranski et al. (2007) <sup>22</sup>   | N = 64 participants formed 16 teams of n = 4. N = 8 teams from Military reserve units and n = 8 civilian teams.   | Decision-making                                                           | Team and individual threat assessments of a military surveillance task (TITAN)         | EEG, oral temperature and Stanford sleepiness scores                                                                                  | Sleep loss had a negative effect on decision-making accuracy and decision time. This performance loss associated with fatigue was mediated by being part of a team.                                                                                               | Designing team and task structures' interdependency (or the perception of interdependency) may provide a useful countermeasure when fatigued.                                          |
| Beckner et al. (2021) <sup>6</sup>     | N = 54 currently serving US military members. 5-day SMOS.                                                         | Executive function                                                        | The 10-item Cognition test battery: MP, VOLT, F2B, AM, LOT, ERT, MRT, DSST, BART, PVT. | CD-RISC, aerobic fitness, blood samples (α-klotho, BDNF, IGF-1, Oxytocin, NPV).                                                       | Fitness and resilience may buffer the impact of simulated military operational stress on vigilance.                                                                                                                                                               | The military may benefit by training and/or selection processes targeting at augmenting trait resilience and aerobic fitness for increased readiness.                                  |
| Beer et al. (2017) <sup>30</sup>       | N = 12 active-duty military personnel. Hypoxia study at synthetic workstation.                                    | Memory, visual and auditory monitoring, reaction time                     | Short-term memory, arithmetic, visual and auditory monitoring using SYNWIN.            | GCMS for gas composition, and pulse oximetry                                                                                          | Severe hypoxia generates greater cognitive impairment than moderate conditions. Memory and visual monitoring more robust than mathematical and auditory processing.                                                                                               | Reliance on auditory warnings to alert aircrew of urgent issues could increase the risk of a missed or ignored warning signal.                                                         |
| Bequette et al. (2020) <sup>26</sup>   | N = 12 active duty or reserve officer training corps completed patrol task with/without exoskeleton               | Visual and auditory monitoring                                            | Context specific visual scanning and audio tasks responding to radio calls.            | NASA-TLX, IMU                                                                                                                         | Wearing a powered exoskeleton during a simulated patrol task significantly impaired visual reaction time.                                                                                                                                                         | Perception of increased workload highlights need to consider "cognitive fit" in the design and implementation of exoskeletons.                                                         |
| Bhattacharyya (2017) <sup>27</sup>     | N = 12 infantry soldiers. 10 min load carriage walking trial on sandy vs. hard surfaces.                          | Attention, memory, executive function                                     | CRT, PRM, SSP.                                                                         | Environmental conditions (DBT, WBT, & RH), ambient temperature, HR, and gas analysis for %VO <sub>2</sub> max.                        | Load, terrain (i.e., sandy vs. hard) and the interaction of, did not produce any significant effect (p > 0.05) on the cognitive performance. However, relative workload and attention significantly correlated on hard terrain for 0 kg, 21.4 kg and 30 kg loads. | Carrying loads of up to 30 kg for short periods of time (i.e., 10 min) on sandy terrains may temporarily improve attention and executive function relative to baseline measures.       |
| Bouak et al. (2018) <sup>31</sup>      | N = 16 military helicopter pilots. Simulated hypoxic flight on desktop simulator.                                 | Short-term and working memory, executive function                         | Delayed matching-to-sample test; the adaptive dual n-back test and Stroop task.        | Cerebral and finger pulse oxymoglobin saturation levels, HR, and respiration rates, SSQ, PANAS and fatigue.                           | Target identification accuracy and latency within a simulated flight task showed decrements at all altitudes (8K, 10K, 12K & 14K).                                                                                                                                | Decrement in target identification latency and accuracy at 8K and 10K may negatively impact flight performance.                                                                        |
| Bouak et al. (2019) <sup>32</sup>      | N = 17 military helicopter pilots. Simulated hypoxic flight on desktop simulator.                                 | Short-term memory and working memory, executive function & multi-tasking. | Delayed matching-to-sample test; the adaptive dual n-back test and Stroop task.        | Cerebral and finger pulse oxymoglobin saturation levels, HR, and respiration rates, PANAS and fatigue.                                | Across 6 h simulated helicopter flights (8K & 10K), expected decreases in O <sub>2</sub> saturation were observed. However, no negative decrements observed for cognitive performance.                                                                            | Participants exposed to mild levels of hypoxia without supplemental O <sub>2</sub> for up to 6 h showed no clear evidence of significant performance decrements.                       |
| Dickinson et al. (2016) <sup>17</sup>  | College students: N = 99 from a federal military academy; n = 100 from a regional state university.               | Executive function                                                        | MATB-II assessed four main, concurrent and independent tasks. Bayes' Rule Task         | Actigraphy device, sleep diaries                                                                                                      | Those who were sleep deprived placed less decision weight on new evidence, relative to base rate information, in making difficult choices.                                                                                                                        | Those who are sleep deprived are at a disadvantage if processing new information is important.                                                                                         |
| Hawes et al. (2012) <sup>33</sup>      | N = 24 United States Army infantry members stationed at the Natick Soldier Systems Center as research volunteers. | Visual perception, executive function                                     | Auditory sequence monitoring task, verbal event planning task spatial map study task   | Two color recognition tasks: Pseudoisochromatic plates and the Farnsworth-Munsell 100 color hue test, adapted Snellen Eye chart, POMs | Higher color temperatures led to faster performance, increased arousal states, lower rated depression and faster verbal and spatial reaction times.                                                                                                               | Lighting effects on mood can be predictive of changes in cognitive performance                                                                                                         |
| Kamimori et al. (2015) <sup>24</sup>   | N = 20 Special Forces personnel. 3-nights of sustained wakefulness with SMOS.                                     | Vigilance; executive functioning                                          | PVT, field vigilance task, logical reasoning                                           | Actigraphy, vigilance monitor, live-fire marksmanship test with friend vs. foe discrimination.                                        | FVT and response speed to LRT tests improved across three overnight testing periods of sustained wakefulness without any decrements in PVT speed or changes to live-fire marksmanship with caffeine use.                                                          | Caffeine use (4 × 200 mg/24 h) can sustain overnight cognitive function in occupational settings that provide less than optimal periods of sleep during successive days of operations. |
| Kobbelvedt et al. (2005) <sup>19</sup> | N = 22 infantry soldiers.                                                                                         | Executive function                                                        | A rescue-operation planning task with manipulation                                     | Self-judgment questionnaire, expert questionnaire                                                                                     | Rescue operations is somewhat resilient to adverse effect from sleep deprivation and time pressure.                                                                                                                                                               | Sleep-deprived individuals can perform well on complex cognitive tasks. Cognitive motivation may influence effort in sleep-deprived individuals and consequently affect performance.   |

(continued on next page)



Table 2 (continued)

| Author (year)                         | Participants and context                                                                | Cognitive function                                                                                                             | Cognitive tests                                                                                                                                      | Other measures                                                                                                                                                                                                                     | Key results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Practical considerations                                                                                                                                                                                                                                                         |
|---------------------------------------|-----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lieberman et al. (2002) <sup>8</sup>  | N = 68 Navy SEAL trainees during Hell Week.                                             | Attention, reaction time, short-term working memory and pattern recognition, motor learning                                    | Scanning visual vigilance, four-choice visual reaction time test; matching-to-sample test                                                            | Simulated marksmanship; cortisol, POMs, Stanford sleepiness scale, awareness of caffeine ingestion questionnaire.                                                                                                                  | All cognitive and mood parameters were adversely affected during Hell Week compared to baseline. Caffeine ingestion (200 and 300 mg) significantly improved visual vigilance, choice reaction time, repeated acquisition, self-reported fatigue and sleepiness with the greatest effects on tests of vigilance, reaction time, and alertness.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | When cognitive performance is critical and must be maintained during exposure to severe stress, administration of caffeine may provide a significant advantage. The greatest effects of caffeine were present 1 h post-administration, but significant effects persisted for 8 h |
| Lieberman et al. (2005) <sup>9</sup>  | N = 31 male officers from an elite unit                                                 | Executive function, attention, memory                                                                                          | Four-choice visual reaction time test, scanning visual vigilance test, matching-to-sample test, repeated acquisition test, grammatical reasoning     | POMs, activity and sleep assessment, body composition and hydration status, saliva cortisol, testosterone, and melatonin.                                                                                                          | Sleep loss severely degrades cognitive function, vigilance, reaction time, attention, memory, mood and reasoning.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Highlights the adverse impact of multiple stressors on cognitive performance, mood, and physiologic parameters. This is important as the ability to process information and make correct decisions is critical on the battlefield and other high-stress environments             |
| Lieberman et al. (2006) <sup>11</sup> | N = 13 healthy male soldiers performed 4-day SMOS.                                      | Executive function attention, learning and memory                                                                              | Visual vigilance, four-choice reaction time, pattern recognition, matching to sample test, repeated acquisition test                                 | Actigraphs (for sleep and activity), marksmanship, POMs                                                                                                                                                                            | Significant decrements in visual vigilance, choice reaction time, and matching-to-sample. Cognitive function declined more extensively and rapidly than physical performance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | SMOS could be used to examine treatments that mitigate effects of acute, multi-stressor environments on human performance.                                                                                                                                                       |
| Lieberman et al. (2009) <sup>16</sup> | N = 15 male volunteers from US Army Ranger training                                     | Executive function, attention, memory                                                                                          | Visual vigilance, four-choice reaction time, pattern recognition, symbol–digit substitution, word-list learning, grammatical reasoning               | POMs, Stanford sleepiness scale, rectal temperature, bodyweight, and physiological status                                                                                                                                          | Mood states were adversely affected by training, including tension, depression, anger, fatigue, confusion, and vigor, and acute cold exposure significantly degraded vigilance, overall mood, and increased tension.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Important during disasters and military operations to aim to prevent adverse cognitive and physiological outcomes due to chronic multifactorial stress                                                                                                                           |
| Lieberman et al. (2016) <sup>12</sup> | N = 60 males from the U.S. Army Survival, Evasion, Resistance, and Escape (SERE) school | Language, attention, memory                                                                                                    | Grammatical reasoning, sustained attention, working memory                                                                                           | POMs, heart rate, serum, salivary cortisol and dehydroepiandrosterone-sulfate, serum epinephrine, norepinephrine, and soluble transferrin receptors, salivary neuropeptide-Y, serum prolactin and serum and salivary testosterone. | Cognition, mood, stress hormones, nutritional status and heart rate are simultaneously altered during captivity, and each of these subsequently recovers at different rates. Reasoning, attention, working memory and various aspects of mood were degraded, cortisol, DHEA-s, prolactin and testosterone were significantly lower during captivity. Epinephrine, norepinephrine and neuropeptide-Y and heart rate increased. There were fewer correct responses on the vigilance task when carrying a load or when walking over obstacles. Less distance was covered when carrying a load, traversing the course with obstacles, and when responding to a tactile, compared with an auditory, stimulus. Self-ratings of situational awareness were higher, and not consistent with, expert-observer ratings. Sleep deprivation may impact on self-monitoring and subsequent decision-making | Highlights the severe and unique stresses of simulated capture, interrogation and forced captivity on cognition                                                                                                                                                                  |
| Mahoney et al. (2007) <sup>28</sup>   | N = 18 volunteers from pool of military members. 6 × 30 min load carriage trials.       | Attention and perceptual-motor function                                                                                        | A wrist worn vigilance monitor with auditory, visual, and tactile stimulus. Accuracy and reaction time assessed.                                     | 3 conditions: standing, walking unobstructed, or with small obstacles; loaded (40 kg) vs. unloaded                                                                                                                                 | Performance data from sedentary, rested volunteers may not be relevant to dismounted soldiers performing concurrently physically and cognitively demanding tasks.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                  |
| Matthews et al. (2011) <sup>13</sup>  | N = 30 Norwegian military academy cadets during a summer combat survival course.        | Executive function                                                                                                             | Self-Situation Awareness Behavioral Rating Scale (SABARS) and Expert SABARS.                                                                         | None                                                                                                                                                                                                                               | Self-ratings of situational awareness were higher, and not consistent with, expert-observer ratings. Sleep deprivation may impact on self-monitoring and subsequent decision-making                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Self-assessments of situational awareness may not be accurate under stressful conditions; Researchers must be willing to adapt more established and psychometrically valid SA metrics                                                                                            |
| Morgan et al. (2009) <sup>14</sup>    | N = 132 active-duty U.S. Army soldiers completing survival training.                    | Visual-motor skill, visual perception, visuospatial organization and construction, high-load visual learning and visual memory | The Rey-Osterrieth and Taylor Complex figure copy and recall tasks. DSST; Stroop Task, letter cancellation task, two auditory verbal learning tasks. | 3 different liquid CHO doses or placebo beverage administered at conclusion of the mock captivity phase.                                                                                                                           | Performance on complex cognitive tasks (i.e., Stroop test) was significantly improved with supplemental CHO vs. placebo. No differences were observed on a range of cognitive tasks of lesser complexity.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Supplement strategies may enhance field performance and capacity to assist in sustaining operations in military personnel.                                                                                                                                                       |

|                                       |                                                                                            |                                                  |                                                                                                                                                                                                                         |                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                               |
|---------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Nibbeling et al. (2014) <sup>15</sup> | N = 22 infantry soldiers                                                                   | Higher level, memory, attention, calculation     | Decision and shoot task, shooting accuracy task, math task, memory task, vigilance task.                                                                                                                                | Anxiety thermometer (VAS), rating scale of perceived mental effort, Borgs RPE, and heart rate.                                                                                                                                          | Shooting accuracy and decision-making and mathematical skills decreased significantly under anxiety.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Anxiety can impact performance and exercise-induced fatigue may improve or deteriorate performance in combination with anxiety                                                                                                                                                |
| Owen et al. (2004) <sup>16</sup>      | 18 male subjects during a 4-day combat training military field exercise                    | Memory, attention                                | Cognitive function                                                                                                                                                                                                      | Fasting blood glucose, body mass, mood, activity using wrist-worn activity monitors                                                                                                                                                     | Increase in symptoms relating to hypoglycemia after 24 h in the field. Vigor decreased, and fatigue increased. After 48 h, subjects reported feelings of depression, anger, and confusion. Delayed memory recall was significantly impaired after 48 h, and there was a decrease in vigilance. Between 48 and 72 h, there was a decrease in immediate memory recall. Delayed memory recall and vigilance remained impaired but did not deteriorate further. Degree of sleep deprivation accounted fully for the changes in cognition and well-being. Functional reaction time and peak head velocity increased with equipment load, suggesting reduced perception–action coupling under load. Reduced accuracy and longer firing latencies during marksmanship with increased equipment weight and more challenging postures were also demonstrated. | Highlights that there is a critical period during the first days of field training when cognitive function is most impaired                                                                                                                                                   |
| Palmer et al. (2013) <sup>29</sup>    | N = 6 expert Special Forces soldiers                                                       | Psychomotor attention                            | Reaction time                                                                                                                                                                                                           | Movement, functional reaction time, head turning, firing latency, accuracy                                                                                                                                                              | Participation in a sustained combat course impaired working and long-term memory, but not short-term spatial memory and planning. Cognitive impairments were associated with energy metabolism modulation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Findings highlight that the reductions in functional performance may have significant operational consequences in terms of soldier survivability in combat                                                                                                                    |
| Piérard et al. (2004) <sup>17</sup>   | N = 21 male soldiers on 5-day combat training course                                       | Memory, higher-level, perception,                | Visuomotor Skills, Visuomotor Coordination Test, Selective Attention Test, Planning Test Memory Tests (memory span tests, visual memory test, audiovisual association test, spatial memory tests, semantic memory test) | Sleep and Nutrition Questionnaire, Subjective Vigilance and Mood Testing (visual analog scale), Evaluation of Psychomotor and Cognitive Performance (researchers observing participants complete the tests), blood and saliva sampling. | Visual perceptual, complex motor, and vigilance decrements all occurred with less than 24 h of continuous wakefulness. There was no difference in time of onset of impairments among the three metrics. Significant correlations between NE, total catecholamine levels, and Marine Identity provide further evidence regarding the impact that this training had on the relationship between hormonal responses, cognition, and moral decision-making.                                                                                                                                                                                                                                                                                                                                                                                              | Limited other research like this, however, causal relationships need further examination                                                                                                                                                                                      |
| Russo et al. (2005) <sup>25</sup>     | N = 8 military pilots performed simulated 12.5 h overnight mission in an ARPTT.            | Executive function, attention, motor performance | Choice visual perception task (CVPT), complex motor flight task, and PVT                                                                                                                                                | Actigraphy, ARPTT task performance.                                                                                                                                                                                                     | Visual perceptual, complex motor, and vigilance decrements all occurred with less than 24 h of continuous wakefulness. There was no difference in time of onset of impairments among the three metrics.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Supports the use of visual perceptual measures in place of complex motor performance measures, in situations where the individual is dependent on primarily visual information                                                                                                |
| Siedlik et al. (2016) <sup>18</sup>   | N = 7 active-duty male Marines during a typical Marine Corps Martial Arts training session | Higher-level, emotion                            | Assessments of cognitive moral functioning (moral judgment and intent)                                                                                                                                                  | Blood samples, the Spielberger State-Anxiety Inventory, Moral Identity Instrument, Marine Identity and National Identity, the moral functioning continuum.                                                                              | Significant correlations between NE, total catecholamine levels, and Marine Identity provide further evidence regarding the impact that this training had on the relationship between hormonal responses, cognition, and moral decision-making.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | This research has identified variations in cognitive function that suggest a link between physiological parameters and moral decision-making. A better understanding of this link can aid in the design of interventions to improve the individual's decision-making ability. |

(continued on next page)

Table 2 (continued)

| Author (year)                           | Participants and context                                                                                     | Cognitive function                         | Cognitive tests                                                                                                                                                                                                                                                                                                                                                 | Other measures                                                                                                                                                                                                                                                                                               | Key results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Practical considerations                                                                                                                                                |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Smith et al. (2019) <sup>2</sup>        | N = 20 active-duty male soldiers. 72 h with sleep restriction.                                               | Attention, working memory, decision-making | Go/No-Go task, and a visual n-back task.                                                                                                                                                                                                                                                                                                                        | Friend v. foe marksmanship discrimination task with varying levels of cognitive load; and a modified Army Record Fire task. NASA-TLX                                                                                                                                                                         | Over a 72-h period with only 2 h of sleep per night, soldiers take significantly longer to make decisions, make the wrong decisions regarding friend or foe identification, and do not believe their performance is suffering. Scores on all measures were degraded during Conduct After Capture training, but recovered after completion of training, and almost all measures were most degraded at the more intense interrogation role-play scenario. Psychomotor speed, reaction time, visual tracking and vigilance were significantly impaired following training with > 24 h sleep deprivation. The majority of the affected measures recovered within two days of leaving the field training environment, despite continuing pickets in a simulated forward operating base. | Results characterize the consequences of sleep restriction on marksmanship and reinforce the importance of adequate sleep among service members when feasible.          |
| Suurd Ralph et al. (2017) <sup>19</sup> | N = 36 (1 female) Canadian Armed Forces military personnel undergoing a 4-day captivity survival course.     | Higher-level, memory, attention            | Delayed Matching-to-Sample, n-back                                                                                                                                                                                                                                                                                                                              | Brief Trauma Questionnaire, childhood trauma questionnaire, Clinician-Administered Dissociative States Scale, PTSD Symptom Scale, Multidimensional Fatigue Inventory, Profile of Mood States, wrist actigraphy Sleep (24 h actigraphy), subjective workload (NASA-TLX), fatigue (Samm-Perelli Fatigue Scale) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Provides evidence to suggest that captivity survival training impacts psychological and physiological function.                                                         |
| Tait et al. (2022) <sup>20</sup>        | N = 57 (3 female) soldiers undergoing the Australian Army combat engineer Initial Employment Training Course | Memory, vigilance, attention               | The Motor Praxis Task (MPT): psychomotor speed and visual tracking; Visual Object Learning Task (VOLT): working memory and visual object learning; Line Orientation Task (LOT): visuospatial orientation; Digit Symbol Substitution Task (DSST): attention, working memory, complex scanning and visual tracking; PVT: psychomotor speed and vigilant attention |                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | The sensitivity of the cognitive tests to sleep restriction, and recovery, indicates their potential utility to assess operational readiness in military personnel.     |
| Trousselard et al. (2009) <sup>21</sup> | N = 13 qualified submariners. Escape from sunken submarine vs. simulation task                               | Memory                                     | Declarative memory; digit span test                                                                                                                                                                                                                                                                                                                             | ECC, cortisol, POMs and sleep                                                                                                                                                                                                                                                                                | Escape at sea showed significant impairment in declarative memory, higher cortisol and adverse changes in mood compared to the equivalent simulated exercise in the tank.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | The psychophysiological and cognitive effects of military exercises may be influenced by the realism of conditions and by prior exposure to life-threatening situations |

Note: AM = abstract matching; ARPTT = Air Refueling Part Task Trainer; BART = balloon analog risk test; BDNF = brain-derived neurotrophic factor; CD-RISC = Connor-Davidson Resilience Scale; CRT = choice reaction time; DBT = dry bulb temperature; DSST = digit symbol substitution test; EEG = electroencephalogram; ERT = emotion recognition test; F2B = Fractal 2-Back; GCMS = gas chromatograph mass spectrometer; HR = heart rate; ICF-I = insulin-like growth factor I; IMU = inertial measurement units; LOT = line orientation test; MP = motor practice test; MRT = matrix reasoning test; NPY = neuropeptide Y; PANAS = positive and negative affects scale; POMs = profile of mood states questionnaire; PRM = pattern recognition memory; PTSD = post traumatic stress disorder; PTV = Psychomotor Vigilance Task; RH = relative humidity; SMOS = simulated military operational stress; SSP = spatial span; SSQ = signs and symptoms questionnaire; SYNWIN = a synthetic computer workstation in which the four tasks are presented simultaneously; VOLT = visual object learning test; WBT = wet bulb temperature.

Therefore, the ability to monitor cognitive function with the goal of managing allostatic load may facilitate efforts to optimize performance, resilience, and long-term health of military personnel.

Language-, attention-, and memory-related cognitive functions were all degraded following simulated captivity and mock interrogations.<sup>12,19</sup> Army ranger training was associated with decrements in attention and higher-level cognitive functions, and decrements in sustained attention were noted after acute cold exposure.<sup>16</sup> Furthermore, Matthews and colleagues<sup>13</sup> demonstrated that the ability to rate situational awareness is compromised during periods of extreme stress (food and sleep deprivation). These latter findings suggest that self-assessments of cognitive function may be unreliable when accuracy is most important (i.e., during high stress situations), and thus underscore the importance of objective measurement and mitigation techniques.

There were a number of instances where participants were exposed to complete sleep deprivation<sup>7,13,22</sup> or restriction (i.e. ~4 h)<sup>2,9,11,17,20,24,25</sup> for the duration or majority of the data collection period. Decrement in cognitive function were apparent in all of these cases, with one exception; during simulated search and rescue training,<sup>7</sup> military cadets that had been exposed to 52–100 h sleep deprivation displayed no impairments in cognitive function when there was a 'need for cognition' in the planning of a rescue operation. While the possible influence of other transient compensatory adaptations must be considered, the results suggest that procedural knowledge was resilient to sleep deprivation and time pressure. Thus, various psychophysiological processes may allow sufficiently trained individuals to temporarily buffer some of the negative effects of sleep restriction or deprivation, but this relationship appears complex and multifaceted. Over time, repeated exposure to such situations (with inadequate periods of recovery) would be expected to degrade cognitive function.

In addition to the impact of sleep loss on cognitive function, this review confirmed that other stressors (e.g., physical exertion,<sup>11</sup> caloric restriction,<sup>13</sup> hypoxia,<sup>30–32</sup> psychological stress,<sup>19</sup> thermal stress<sup>9,11</sup>) influence cognitive function in military populations. We have attempted to collate the evidence to create a foundation for future efforts to assess cognitive function in this context. The tests reviewed in this paper cover a range of cognitive domains including higher-level executive functions (e.g. planning, judgment, flexibility, problem-solving), plus attention, perception, and memory. While most of these tests are computer-based, some can be administered with pen and paper, mobile phone, or tablet and thus facilitate implementation in field settings. Although more work needs to be done, the wearable vigilance monitor that has been developed by Lieberman et al.<sup>34</sup> may present as an attractive alternative to the computer-based tests to monitor cognitive function in military settings.

Not captured by the current review is the work that explores how declines in cognitive function can place the musculoskeletal system at risk; an important area for future research. Marksmanship judgment and accuracy are negatively affected by cognitive fatigue associated with repetitive cognitive<sup>35</sup> and prolonged load carriage tasks,<sup>36</sup> respectively. In addition to occupational tasks with substantial cognitive demands, task-related decreases in cognitive function can impair more basic aspects of musculoskeletal function.<sup>37,38</sup> In this case, decreases in function are related to perceptions of fatigue, cognitive or physical demands, and resources.<sup>39</sup> Rather than a generalized loss of function, musculoskeletal impairments appear greatest for endurance-type activities and may depend on task duration and intensity.<sup>40</sup> In athletic populations, cognitive function is inversely related to musculoskeletal injury incidence.<sup>41,42</sup> For example, worse reaction times, processing speeds, and memory performance predict non-contact anterior cruciate ligament sprains, a common military injury.<sup>43</sup>

Overall, this review highlights the concept that military personnel encounter multiple stressors, and these stressors can negatively affect cognitive function. In turn, decreases in cognitive function have deleterious effects on occupational task performance, musculoskeletal function, injury risk, and recovery. The stress associated with sleep loss,

sustained cognitive and physical exertion, and psychophysiological aspects of military occupational tasks rarely occur in isolation.<sup>9,12,15,16</sup> Accordingly, interactions among these stressors likely produce compounding effects. Integrative work is needed to better understand the nature of decreases in cognitive function to mitigate its consequences. In this case, the concurrent use of multiple methods (e.g. cognitive tests, inertial measurement units, surveys, and physiological or blood assays) may facilitate immediate and longitudinal assessment of cognitive function, capacity, and fatigue in field settings.

## 5. Practical considerations

There are a number of cognitive functions that are degraded through exposure to military stressors and compromised by sustained operations with restricted sleep. Whether it is appropriate to measure cognitive function during sustained operations is questionable, noting the operational needs, associated cognitive burden and mission imperatives. However, it may be advantageous to monitor the cognitive function of personnel before and following operational activities to better understand the capacity of the team prior to and following a mission. There may also be opportunities to include measures of cognitive function during military training exercises and simulation scenarios to better understand the potential impact of the SMOS environment on cognitive function; and use it as an opportunity to educate personnel after the fact on the possible risks of decline in cognitive function irrespective of individual perceptions of performance maintenance. When cognitive function is critical and must be maintained during exposure to severe stress,<sup>8</sup> or over sustained operations with restricted sleep,<sup>24</sup> administration of caffeine or carbohydrate beverages<sup>14</sup> may provide a significant advantage.

Most studies in this review used a mixture of cognitive tests in conjunction with other measures such as self-report questionnaires, physiological data (i.e., blood or saliva samples for various biomarkers), and actigraphy (for the measurement of physical activity or sleep). Due to the limited research published in the last two years, and the recent advances in our knowledge of the available suite of circulating biomarkers,<sup>44</sup> future research efforts would benefit from the inclusion of these more invasive measures (e.g. blood) in conjunction with non-invasive methods to determine the minimal combination of measures that best reflect psychophysiological responses to common high stress military situations. Given proper integration, this combination of tests may facilitate efforts to assess various elements of cognitive function in complex environments with multiple stressors. Nevertheless, while sensitive to changes in psychophysiological state, many of these biomarkers may possess limited utility in field settings with the technology currently available. For example, the collection of blood and saliva samples can be invasive, costly, and logistically challenging (i.e. phlebotomy, equipment, and storage requirements). Similarly, while there are a number of commercially available wearable devices, depending on the platform, data may not be relatable to measures of cognitive function. For example, actigraphy devices may be worn in the field, but the data need to be downloaded, processed, and (potentially) integrated with other measures before insights on sleep status or physical workload can be obtained.

## 6. Limitations

There were some limitations in the papers reviewed that should be noted. For instance, the majority of studies had small samples, adopted within-subject designs, and involved only male participants. In addition, most studies were published before 2018, with only five studies published in the last five years (i.e., one per year).<sup>6,20,26,31,32</sup> There also remains a considerable gap in terms of practical efforts to monitor warfighter performance in deployment settings. Nevertheless, these studies from SMOS confirm that cognitive tests can provide useful information about cognitive function and fatigue in military personnel. In



addition to non-invasiveness and automation, some of these measures could be incorporated into existing workflows in field settings where there is a need to monitor the prospective performance capacity of the warfighter. Although there were several studies with parallel measures of performance and cognitive tests before, during, or after field training activities, more research is needed to determine and optimize feasibility, acceptability, and fidelity in field settings. Steps to achieve these goals could begin with assessments of larger cohorts during training, and perhaps over time, different training scenarios. In addition to longitudinal data, with a sufficiently large sample size, such efforts would enable more sophisticated data analytics to better understand the usefulness of these measures as biomarkers of cognitive function. The development and validation of context-specific cognitive tests that do not depend on technology may also be a valuable consideration for certain military environments.

## 7. Conclusion

In spite of these limitations, measures of cognitive function represent an efficient way to non-invasively obtain useful information about warfighter health, recovery, and readiness in military operational environments. Although there is a substantial amount of research on cognitive function in the warfighter, data from deployed or operational environments is limited. This review identified several categories of cognitive function tests that have been used in military training scenarios. For example, measures of memory and attention revealed compromised cognitive function after exposure to multi-stressor training environments. This paper serves as a platform for future conversations around the development of ecologically valid cognitive tests that can be used in deployment settings. These measures will need to detect changes in cognitive function reliably and validly and do so in military populations and environments. Once sensitive and reliable measures of cognitive function are established, it will be possible to develop monitoring systems to manage fatigue and optimize performance capacity.

## Funding information

No financial assistance was provided for this project.

## Confirmation of ethical compliance

Ethics approval was not required for this paper.

## Declaration of interest statement

None.

## Transparency declaration

This article is published as part of a Supplement entitled 'Macquarie University [MQ] Military Biomarker Symposium 2022' supported by the Defence Science and Technology Group (DSTG). Funding for the Symposium was provided by Office of Naval Research Global through the Collaborative Science Program. Additional funding was provided by the NSW Defence Innovation Network, Smartabase/Fusion Sport, and Vicon.

## References<sup>1</sup>

- Rice VJB, Schroeder PJ. Self-reported sleep, anxiety, and cognitive performance in a sample of U.S. Military active duty and veterans. *Mil Med* 2019;184(Suppl 1):488-497.
- \*Smith CD, Cooper AD, Merullo DJ et al. Sleep restriction and cognitive load affect performance on a simulated marksmanship task. *J Sleep Res* 2019;28(3):e12637.
- Grandou C, Wallace L, Fullagar HHK et al. The effects of sleep loss on military physical performance. *Sports Med* 2019;49(8):1159-1172.
- Grier T, Dinkeloo E, Reynolds M et al. Sleep duration and musculoskeletal injury incidence in physically active men and women: a study of U.S. Army Special Operation Forces soldiers. *Sleep Health* 2020;6(3):344-349.
- Devine JK, Collen J, Choynowski JJ et al. Sleep disturbances and predictors of nondeployability among active-duty Army soldiers: an odds ratio analysis of medical healthcare data from fiscal year 2018. *Mil Med Res* 2020;7(1):10.
- \*Beckner et al. Impact of simulated military operational stress on executive function relative to trait resilience, aerobic fitness, and neuroendocrine biomarkers. *Physiol Behav* 2021;2021(236):113413.
- \*Kobbeltvedt T, Brun W, Laberg JC. Cognitive processes in planning and judgements under sleep deprivation and time pressure. *Organ Behav Hum Decis Process* 2005;98(1):1-14.
- \*Lieberman HR, Tharion WJ, Shukitt-Hale B et al. Effects of caffeine, sleep loss, and stress on cognitive performance and mood during U. S. Navy SEAL training. *Psychopharmacology* 2002;164(3):250-262.
- \*Lieberman HR, Bathalon GP, Falco CM et al. Severe decrements in cognition function and mood induced by sleep loss, heat, dehydration, and undernutrition during simulated combat. *Biol Psychiatry* 2005;57(4):422-429.
- \*Lieberman HR, Niro P, Tharion WJ et al. Cognition during sustained operations: comparison of a laboratory simulation to field studies. *Aviat Space Environ Med* 2006;77(9):929-936.
- \*Lieberman HR, Castellani JW, Young AJ. Cognitive function and mood during acute cold stress after extended military training and recovery. *Aviat Space Environ Med* 2009;80(7):629-636.
- \*Lieberman HR, Farina EK, Caldwell J et al. Cognitive function, stress hormones, heart rate and nutritional status during simulated captivity in military survival training. *Physiol Behav* 2016;165:86-97.
- \*Matthews MD, Eid J, Johnsen BH et al. A comparison of expert ratings and self-assessments of situation awareness during a combat fatigue course. *Mil Psychol* 2011;23(2):125-136.
- \*Morgan CA, Haslett G, Southwick S et al. Effect of carbohydrate administration on recovery from stress-induced deficits in cognitive function: a double-blind, placebo-controlled study of soldiers exposed to survival school stress. *Mil Med* 2009;174(2):132-139.
- \*Nibbeling N, Oudejans RR, Ubink EM et al. The effects of anxiety and exercise-induced fatigue on shooting accuracy and cognitive performance in infantry soldiers. *Ergonomics* 2014;57(9):1366-1379.
- \*Owen G, Turley H, Casey A. The role of blood glucose availability and fatigue in the development of cognitive impairment during combat training. *Aviat Space Environ Med* 2004;75(3):240-246.
- \*Piérard C, Béracochéa D, Pérès M et al. Declarative memory impairments following a military combat course: parallel neuropsychological and biochemical investigations. *Neuropsychobiology* 2004;49(4):210-217.
- \*Siedlik JA, Deckert JA, Clopton AW et al. Immunoendocrine alterations following Marine Corps Martial Arts training are associated with changes in moral cognitive processes. *Physiol Behav* 2016;154:76-82.
- \*Suurd Ralph C, Vartanian O, Lieberman HR et al. The effects of captivity survival training on mood, dissociation, PTSD symptoms, cognitive performance and stress hormones. *Int J Psychophysiol* 2017;117:37-47.
- \*Tait JL, Aisbett B, Corrigan SL et al. Recovery of cognitive performance following multi-stressor military training. *Hum Factors* 2022. 00187208221086686.
- \*Troussellard M, Cian C, Barraud PA et al. Physiological and psychological effects of escape from a sunken submarine on shore and at sea. *Aviat Space Environ Med* 2009;80(10):850-860.
- \*Baranski JV, Thompson MM, Lichacz FJM et al. Effects of sleep loss on team decision making: motivational loss of motivational gain? *Hum Factors* 2007;49(4):646-661.
- \*Dickinson DL, Drummond SP, Dyche J. Voluntary sleep choice and its effects on Bayesian decisions. *Behav Sleep Med* 2016;14(5):501-513.
- \*Kamimori GH, McLellan TM, Tate CM et al. Caffeine improves reaction time, vigilance and logical reasoning during extended periods with restricted opportunities for sleep. *Psychopharmacology* 2015;232(12):2031-2043.
- \*Russo MB, Kendall AP, Johnson DE et al. Visual perception, psychomotor performance, and complex motor performance during an overnight air refueling simulated flight. *Aviat Space Environ Med* 2005;76(7, Sect2, Suppl):C92-C104.
- \*Bequette B, Norton A, Jones E et al. Physical and cognitive load effects due to a powered lower-body exoskeleton. *Hum Factors* 2020;62(3):411-424.
- \*Bhattacharyya D, Pal M, Chatterjee T et al. Effect of load carriage and natural terrain conditions on cognitive performance in desert environments. *Physiol Behav* 2017;179:253-262.
- \*Mahoney C, Hirsch E, Hasselquist L et al. The effects of movement and physical exertion on soldier vigilance. *Aviat Space Environ Med* 2007;78(5, Sect II, Suppl):B51-B58.
- \*Palmer CJ, Bigelow C, Van Emmerik REA. Defining soldier equipment trade space: load effects on combat marksmanship and perception-action coupling. *Ergonomics* 2013;56(11):1708-1721.
- \*Beer JMA, Shender BS, Chauvin D et al. Cognitive deterioration in moderate and severe hypobaric hypoxia conditions. *Aerospace Med Human Perform* 2017;88(7):617-627.
- \*Bouak F, Vartanian O, Hofer K et al. Acute mild hypoxic hypoxia effects on cognitive and simulated aircraft pilot performance. *Aerospace Med Human Perform* 2018;89(6):526-536.
- \*Bouak F, Vartanian O, Hofer K. Performance and health effects of mild hypoxic hypoxia in simulated 6-hour exposures between 2,438 and 3,048 metres. *J Mil Vet Fam Health* 2019;5(2):40-50.

<sup>1</sup> \* indicates the 29 papers included in the review.

33. \*Hawes BK, Bruny  TT, Mahoney CR et al. Effects of four workplace lighting technologies on perception, cognition and affective state. *Int J Ind Ergon* 2012;42(1):122–128.
34. Lieberman HR, Kramer FM, Montain SJ et al. Field assessment and enhancement of cognitive performance: development of an ambulatory vigilance monitor. *Aviat Space Environ Med* 2007;78(5, Sect II, Suppl):B268–B276.
35. Head J, Tenan MS, Tweedell AJ et al. Prior mental fatigue impairs marksmanship decision performance. *Front Physiol* 2017;8:680.
36. Tenan MS, LaFiandra ME, Ortega SV. The effect of soldier marching, rucksack load, and heart rate on marksmanship. *Hum Factors* 2017;59(2):259–267.
37. Duncan MJ, Fowler N, George O et al. Mental fatigue negatively influences manual dexterity and anticipation timing but not repeated high-intensity exercise performance in trained adults. *Res Sports Med* 2015;23(1):1–13.
38. Marcora SM, Staiano W, Manning V. Mental fatigue impairs physical performance in humans. *J Appl Physiol* 2009;106(3):857–864.
39. Martin K, Meeusen R, Thompson KG et al. Mental fatigue impairs endurance performance: a physiological explanation. *Sports Med* 2018;48(9):2041–2051.
40. Van Cutsem J, Marcora S, De Pauw K et al. The effects of mental fatigue on physical performance: a systematic review. *Sports Med* 2017;47(8):1569–1588.
41. McDonald AA, Wilkerson GB, McDermott BP et al. Risk factors for initial and subsequent core or lower extremity sprain or strain among collegiate football players. *J Athl Train* 2019;54(5):489–496.
42. Wilke J, Groneberg DA. Neurocognitive function and musculoskeletal injury risk in sports: a systematic review. *J Sci Med Sport* 2022;25(1):41–45.
43. Swanik CB, Covassin T, Stearne DJ et al. The relationship between neurocognitive function and noncontact anterior cruciate ligament injuries. *Am J Sports Med* 2007;35(6):943–948.
44. Beckner ME, Main LC, Tait JL et al. Circulating biomarkers associated with performance and resilience during military operational stress. *Eur J Sport Sci* 2021;22(1):72–86.